

# From Behaviorism to Language Models: A Formal Correspondence Between the Matching Law and Token Prediction

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## Abstract

We propose a formal correspondence between principles of operant behaviorism and the optimization dynamics of large language models (LLMs). The matching law, a cornerstone of behavior analysis, states that organisms allocate responses to alternatives in proportion to reinforcement rates. We demonstrate that this law parallels maximum likelihood estimation in LLM pretraining, generalized matching in reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO), and temperature scaling at inference (Guo et al., 2017). Our framework is descriptive rather than normative: it identifies lawful allocation patterns across biological and artificial systems and provides testable predictions that move debate beyond metaphor.

## 1 Introduction

The rise of large language models (LLMs) has intensified debate about what, if anything, these systems learn. Some critics describe them as hightech plagiarism software or stochastic parrots,<sup>1</sup> emphasizing surface imitation over principle (Bender et al., 2021; Chomsky et al., 2023). Others argue that LLMs demonstrate forms of generalization and adaptation that parallel long-standing behavioral and cognitive theories. What is missing is a formal framework that connects the optimization principles underlying LLMs with the empirically validated laws of behavior established in psychology.

In a different intellectual tradition, behavior analysis has long emphasized lawful relations between environmental contingencies and the allocation of behavior. One of the most influential contributions of B.F. Skinner’s radical behaviorism was the analysis of *selection by consequences*: behavior is shaped and maintained by the contingencies of reinforcement in the environment (Skinner, 1953). This perspective led to quantitative laws, most prominently Herrnstein’s matching law, which describes how organisms allocate responses to alternatives in direct proportion to reinforcement rates (Herrnstein, 1961). Subsequent work generalized this principle to account for systematic deviations in sensitivity and bias (Baum, 1974). Subsequent work generalized this principle to account for systematic deviations in sensitivity and bias (Baum, 1974). The matching law continues to be actively investigated and applied, from clinical interventions (Kronfli et al., 2021), to human communication behavior (Anderson, 2024), to neuroscience perspectives on reward expectations (Rajagopalan et al., 2023).

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<sup>1</sup>The phrase “stochastic parrot” is in some sense more literally apt for finite-order  $n$ -gram generators such as Emacs `dissociated-press` or the *Travesty* program published in *Byte Magazine* (Kenner and O’Rourke, 1984). Our results show that contemporary LLMs instantiate allocation laws that go beyond this baseline, though they do not by themselves resolve questions of semantic grounding or bias.

Modern LLMs are trained and aligned by equally precise optimization principles. During pretraining, models minimize crossentropy and therefore perform maximumlikelihood estimation (MLE) to match empirical token distributions. During alignment, objectives such as RLHF and direct preference optimization (DPO) add a reward signal and a KL regularization term against a reference model (Christiano et al., 2017; Ouyang et al., 2022; Rafailov et al., 2023). At inference, decoding procedures such as temperature sampling rescale probabilities in a controlled way. Although it is mathematically straightforward that idealized MLE recovers the data distribution, interpreting this fact through the lens of the matching law yields a cross-disciplinary insight: LLM token probabilities can be treated as allocations of responses under contingencies. This characterization does not anthropomorphize models; it identifies lawful allocation dynamics that are substrate-independent (we refer to this as *allocation-grounding*). In effect, the corpus serves as the models experiential channel: the structured set of contingencies from which allocations are learned. We later describe this as a form of *gestalt-grounding*, by analogy to the sensory gestalt that shapes animal learning (see Discussion for expansion upon this terminology of grounding types applicable to machine learning).

Although reinforcement learning as a field has historical roots in psychology and behaviorism (Sutton and Barto, 1998), the specific objectives used in modern LLM training and alignment have been justified statistically rather than behavior-analytically; to our knowledge, no prior work has shown their direct formal equivalence to matching-law principles.

**Contributions.** We formalize and prove the correspondences between: (1) MLE and strict matching; (2) KLregularized preference optimization and generalized matching with interpretable sensitivity and bias; and (3) temperature scaling and sensitivity rescaling. We outline empirical tests and provide synthetic demonstrations. The framework is descriptive, not normative: it clarifies how allocation behaves under current training regimes, and it supplies testable predictions that move discussion beyond metaphor.

At one level, the correspondences we present may seem almost obvious: learning theory should apply whether the learner is a pigeon, a primate, or a program. Yet, to our knowledge, prior discussions have relied largely on metaphor or analogy rather than on formal correspondence. Our contribution is to replace metaphor with proof: maximum-likelihood estimation instantiates strict matching, KL-regularized alignment instantiates generalized matching, and temperature scaling instantiates sensitivity rescaling. By showing that these well-known optimization principles in machine learning are mathematically equivalent to empirical laws in behavior analysis, we provide a framework that is at once descriptively precise and empirically testable. This formalization, rather than the intuition alone, is what makes the correspondence a novel advance.

## 2 Related Work

Skinner’s *Science and Human Behavior* (Skinner, 1953) articulated a program in which reinforcement schedules and contingencies were sufficient to explain behavioral allocation without recourse to internal cognitive mechanisms. Building on this foundation, Herrnstein (Herrnstein, 1961) introduced the matching law, which formalized the observation that organisms allocate responses in proportion to the relative rates of reinforcement. Baum (Baum, 1974) extended this into the generalized matching law, in which deviations from strict proportionality could be captured by parameters representing sensitivity and bias. Subsequent work in quantitative behavior analysis (e.g., Shimp, 1976; Grace, 1994; Killeen, 1994) incorporated temporal dynamics, signal detection principles, and memory limitations. Despite this rich history, these principles have rarely been applied outside of biological behavior.

Associative models such as Rescorla and Wagner (Rescorla and Wagner, 1972) provided mathematical descriptions of classical conditioning. Later, reinforcement learning frameworks (Sutton & Barto, 1998) aligned these models with both machine learning algorithms and neural mechanisms of reward. Neuroscientific studies have shown that midbrain dopamine neurons encode reward prediction errors in ways consistent with operant formulations. Yet these literatures have remained largely confined to animals and humans, without systematic exploration of their relevance to large-scale language models.

Recent advances in artificial intelligence emphasize the role of scaling laws. Kaplan et al. (Kaplan et al., 2020) demonstrated that model performance follows predictable power-law relationships with respect to parameters, data, and compute. At larger scales, qualitative changes emerge: Wei et al. (Wei et al., 2022) documented emergent abilities such as zero-shot translation and reasoning. These studies highlight the predictive value of statistical regularities at scale but do not ground LLM behavior in pre-existing behavioral science frameworks.

Critical voices have questioned whether LLMs should be considered learners in any meaningful sense. Bender et al. (Bender et al., 2021) described LLMs as stochastic parrots, generating outputs without understanding. Marcus (Marcus, 2022) has emphasized brittleness, lack of reasoning, and susceptibility to failure modes. Chomsky, Roberts, and Watumull (Chomsky et al., 2023) characterized LLMs as mere plagiarism systems, incapable of genuine insight. These critiques highlight the need for principled frameworks to describe what LLMs are doing, beyond metaphor or dismissal.

**Relation to prior links with psychology.** Reinforcement learning as a field has long drawn inspiration from behavioral and cognitive psychology (e.g., Thorndikes law of effect, Skinners operant conditioning), and this lineage is acknowledged in foundational texts (Sutton and Barto, 1998). However, the specific objectives that now underlie large language model training and alignment—maximum likelihood estimation, KL-regularized reinforcement learning from human feedback (Christiano et al., 2017; Ouyang et al., 2022), direct preference optimization (Rafailov et al., 2023), and temperature scaling (Guo et al., 2017)—have been justified in the machine learning literature primarily as statistical or probabilistic principles. To our knowledge, no prior work has established a direct formal correspondence between these objectives and the empirically validated laws of allocation from behavior analysis, such as Herrnsteins matching law (Herrnstein, 1961) or Baums generalized matching law (Baum, 1974). Our contribution is to make this correspondence explicit and provable, replacing metaphor with formal equivalence.

Beyond its historical roots, the matching law remains a live area of research in applied and theoretical contexts. Recent work has applied generalized matching to evaluate treatment outcomes (Kronfli et al., 2021), modeled operant behavior in neural terms (Rajagopalan et al., 2023), and extended allocation analysis to everyday human activities such as texting (Anderson, 2024). Professional practice in applied behavior analysis also continues to highlight matching law principles (Mastermind Behavior, 2025).

The present contribution is the first to systematically map behavior-analytic principles—strict and generalized matching, sensitivity, bias, and contextual constraints—onto the training and inference of LLMs. By framing LLM outputs in terms of the matching law, we re-situate their behavior as part of a broader class of allocation phenomena observed in biological organisms. This does not resolve all questions about generalization, compositionality, or emergent abilities, but it does provide a benchmark descriptive framework with which to analyze LLM performance.

### 3 Formal Correspondence

We summarize the main theoretical correspondences and provide statements sufficient for empirical testing.

#### 3.1 Notation and Setup

Let  $c$  denote contexts (prompts) and  $y$  responses (tokens). Let  $q(y | c)$  be the empirical next-token distribution, and  $\pi_\theta(y | c)$  the model policy. The MLE loss is

$$\mathcal{L}_{\text{MLE}}(\theta) = \mathbb{E}_{(c,y) \sim q}[-\log \pi_\theta(y | c)] = \mathbb{E}_{c \sim q}[H(q(\cdot | c), \pi_\theta(\cdot | c))].$$

For KL-regularized RLHF/DPO at a fixed  $x$ , define

$$\mathcal{J}(\pi) = \mathbb{E}_{y \sim \pi(\cdot | x)}[r(x, y)] - \beta \text{KL}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)).$$

#### 3.2 Theorem 1: Strict Matching under MLE

**Statement.** If  $q(\cdot | c)$  is representable, any global minimizer satisfies  $\pi^*(\cdot | c) = q(\cdot | c)$ ; hence for  $a, b$ :

$$\frac{\pi^*(a | c)}{\pi^*(b | c)} = \frac{q(a | c)}{q(b | c)}.$$

**Sketch.**  $H(q, \pi) = H(q) + \text{KL}(q \| \pi)$  is minimized iff  $\pi = q$ .

#### 3.3 Theorem 2: Generalized Matching under KL-RLHF/DPO

**Statement.** The optimizer of  $\max_\pi \mathbb{E}_\pi[r] - \beta \text{KL}(\pi \| \pi_{\text{ref}})$  is

$$\pi^*(y | x) \propto \pi_{\text{ref}}(y | x) \exp\{\beta r(x, y)\}.$$

Therefore

$$\log \frac{\pi^*(y_1 | x)}{\pi^*(y_2 | x)} = \log \frac{\pi_{\text{ref}}(y_1 | x)}{\pi_{\text{ref}}(y_2 | x)} + \beta [r(x, y_1) - r(x, y_2)].$$

This realizes generalized matching with bias from  $\pi_{\text{ref}}$  and sensitivity  $s = \beta$ .

#### 3.4 Theorem 3: Temperature Scaling

Sampling with temperature  $T > 0$  rescales logits, giving  $\pi_T(y | x) \propto \pi(y | x)^{1/T}$  and thus

$$\log \frac{\pi_T(y_1 | x)}{\pi_T(y_2 | x)} = \frac{1}{T} \log \frac{\pi(y_1 | x)}{\pi(y_2 | x)}.$$

#### 3.5 Theorem 4: Mixtures and Regularization

If  $q = \sum_k \alpha_k q^{(k)}$ , then under capacity  $\pi^* = q$  and pairwise odds follow mixture odds. Regularizers that shrink logits (KL to  $\pi_{\text{ref}}$ , label smoothing, entropy bonuses) attenuate log-odds (undermatching).

## 4 Methods

We describe empirical tests that operationalize the theoretical correspondences using standard corpora and openweight LLMs.

**Corpusmodel matching (strict matching).** Select contexts  $c$  from a heldout corpus (e.g., Wikipedia) and candidate continuations  $a, b$  with sufficient counts. Estimate  $\hat{q}(y \mid c)$  by relative frequencies and query pretrained models for  $\log \pi(y \mid c)$ . Regress  $\log \frac{\pi(a|c)}{\pi(b|c)}$  onto  $\log \frac{\hat{q}(a|c)}{\hat{q}(b|c)}$ . Prediction: slope  $s \approx 1$ , intercept  $\alpha \approx 0$ ; finitecapacity and finitesample effects produce systematic undermatching.

**Alignment tuning (generalized matching).** Starting from a base model and reference policy  $\pi_{\text{ref}}$ , perform DPO or PPOstyle RLHF at a range of KL weights  $\beta$ . For preference pairs  $(y_1, y_2)$  given fixed  $x$ , theory predicts

$$\log \frac{\pi(y_1 \mid x)}{\pi(y_2 \mid x)} = \log \frac{\pi_{\text{ref}}(y_1 \mid x)}{\pi_{\text{ref}}(y_2 \mid x)} + s [r(x, y_1) - r(x, y_2)],$$

with  $s = \beta$ . Estimate  $s$  and test linearity and goodnessoffit.

**Temperature scaling.** For a fixed model, vary sampling temperature  $T \in \{0.5, 1, 1.5, 2, 3\}$  and reestimate slopes  $s_T$  in the same logodds regressions. Prediction:  $s_T = s/T$ .

**Mixture corpora and regularization.** When training on mixtures  $q = \sum_k \alpha_k q^{(k)}$ , strict matching implies odds follow mixture odds; verify with small models. Test label smoothing and entropy bonuses, which should attenuate logodds (undermatching).

**Reproducibility.** Report slopes, intercepts,  $R^2$ , and confidence intervals. Release scripts for corpus handling, model queries, and regressions. Synthetic figures here are proofofprinciple; empirical validation on openweight models is planned.

### Synthetic Prototypes

We include synthetic demonstrations to illustrate the correspondence between machine learning objectives and matching-law principles in simplified settings. These prototypes serve two purposes. First, they highlight *allocation-grounding*: the lawful and substrate-independent relation between contingencies and allocations. For example, a toy model minimizing cross-entropy over binary outcomes exhibits strict matching of allocation ratios to reinforcement ratios. Second, they highlight *gestalt-grounding*: the role of the experiential channel in constraining what contingencies are available to the learner. For example, limiting the context window length in a synthetic text model produces systematic undermatching, analogous to perceptual bandwidth limits in animal experiments.

The value of synthetic prototypes lies not in their realism but in their clarity. By controlling all parameters, they make it transparent how allocation laws emerge from optimization objectives, and how deviations can be diagnosed as bias, sensitivity change, or gestalt constraint. This provides a baseline against which both animal behavior and large-scale LLMs can be compared.

#### 4.1 Results: Synthetic Prototypes

## 5 Empirical Program

While our formal correspondences establish a descriptive framework, empirical validation is essential. We therefore outline a program of experiments at two levels: *minimum viable demonstrations* and *optional extensions*.

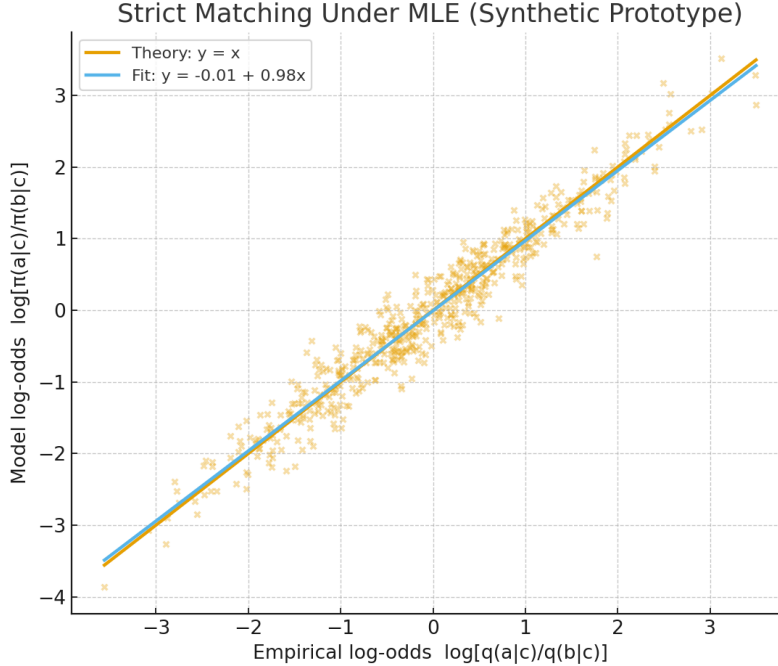


Figure 1: Strict matching under maximum-likelihood pretraining.

## 5.1 Minimum Viable Demonstrations

### 1. Strict Matching Regression

Compute token-level logits from a pretrained model (e.g., GPT-2 small). Compare relative logit ratios for token pairs against empirical corpus frequencies. Demonstrate linear correspondence consistent with Herrnsteins strict matching law.

### 2. Generalized Matching via Alignment

Fine-tune a small model with direct preference optimization (DPO) on a curated preference dataset. Fit generalized matching parameters (sensitivity, bias) to observed allocation shifts. Show that KL-regularization behaves analogously to generalized matching in animal choice.

### 3. Temperature Scaling

Generate outputs from a pretrained model at different sampling temperatures. Fit slopes of allocation curves to verify rescaling of sensitivity.

These three experiments together suffice to demonstrate that reinforcement-allocation principles hold across pretraining, alignment, and inference.

## 5.2 Optional Extensions

- **Scaling Across Model Sizes**

Replicate strict matching regression with models from 124M to 7B parameters. Assess whether sensitivity to corpus statistics improves with scale, paralleling quantitative shifts in animal matching across species.

- **Memory and Context Constraints**

Evaluate strict matching at varying context window lengths. Test the hypothesis that longer

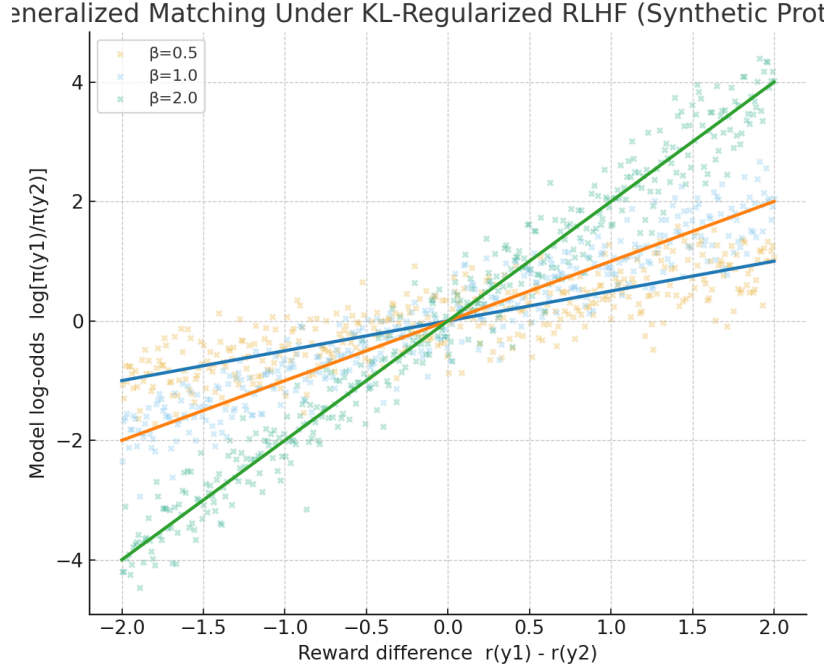


Figure 2: Generalized matching under KL-regularized RLHF/DPO.

windows yield closer adherence to corpus contingencies, paralleling working-memory effects in animals.

- **Comparative Datasets**

Apply the same analyses to corpora with distinct statistical profiles (e.g., natural text vs. synthetic reinforcement schedules). Assess robustness of the correspondence.

### 5.3 Summary

The *minimum program* (strict matching, generalized matching, temperature scaling) establishes the core claim of a formal correspondence. The *optional extensions* provide depth, linking allocation fidelity to model size, memory, and context — and thereby positioning LLMs more explicitly within the comparative framework of behavioral science.

## 6 Broader Research Program

The correspondences we have established are not merely conceptual. They yield testable predictions and point toward an empirical program that bridges behavior analysis and machine learning. This program has two complementary strands: allocation-grounding (substrate-independent laws of matching) and gestalt-grounding (substrate-specific constraints of perception or corpus). The particular contingencies that a system encounters depend on its gestalt—sensory in organisms, textual in LLMs—so allocation-grounding and gestalt-grounding are complementary rather than contradictory perspectives.

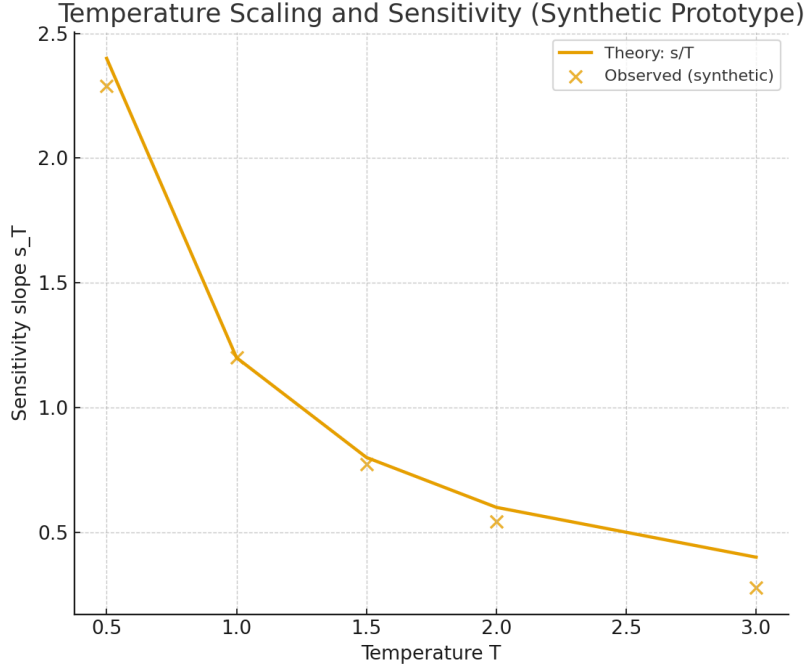


Figure 3: Temperature scaling effects on sensitivity.

## Cross-Cutting Standards

To ensure comparability across models and organisms, we recommend reporting:

- sensitivity (slope) and bias (intercept) from log-odds regressions;
- $R^2$  with confidence intervals;
- sample size, context granularity, and effective context length;
- regularization and decoding parameters (KL weights, temperature, top- $k$ ,  $p$ , repetition penalties);
- corpus domain or sensory channel (textual vs. perceptual gestalt (Koffka, 1935))

## Core Tracks

**Track A: Allocation Benchmarks.** Replicate strict matching (MLE), generalized matching (KL-regularized alignment), and temperature scaling on small LLMs and animal models. *Milestone:* slopes near predicted values with interpretable bias.

**Track B: Capacity and Gestalt.** Vary context window (LLMs) or sensory bandwidth (organisms) to probe gestalt constraints. *Milestone:* undermatching increases with reduced window/bandwidth; sensitivity improves with expanded gestalt.

**Track C: Comparative Phenomena.** Test classical conditioning (Rescorla and Wagner, 1972) and in-context learning (Brown et al., 2020) analogs. *Milestone:* recovery of cueoutcome learning curves, blocking, and observational-learning analogs.

**Track D: Mechanistic Probes.** Investigate how embeddings (LLMs) or neural circuits (organisms) implement allocation adjustments. *Milestone:* correlation between representational change and shifts in sensitivity/bias.

## Program Matrix

Table 1 provides a mapping from behavioral constructs to LLM mechanisms and specifies minimal and stretch experiments. Importantly, each row can be interpreted through both allocation-grounding and gestalt-grounding: allocation laws define what patterns to expect, while gestalt constraints define what contingencies are accessible to the learner.

## Risks and Mitigations

*Data bias:* stratify by domain; *Overfitting to proxies:* validate on held-out corpora and alternate tokenizations; *Temperature/filters:* report full decoding stack; *Comparability:* match stimulus domain to gestalt channel.

## Artifacts and Reproducibility

Release code, prompts, and corpora slices; provide CSVs with per-pair odds, slopes, intercepts, and CIs; include model hashes and, for animal studies, apparatus schematics. This ensures cumulative science across both allocation and gestalt dimensions.

## Summary

The research program moves discussion beyond metaphor by embedding LLMs in the same empirical cycle long used in behavior analysis: specify contingencies, measure allocations, test laws, and interpret deviations. Allocation-grounding supplies the general lawfulness; gestalt-grounding specifies the substrate that enables or constrains it.

## 7 Discussion

The formal results show that core principles of operant psychology apply to LLM behavior at the level of allocation. Pretraining under MLE yields strict matching in the idealized limit; alignment with KLregularized reward yields generalized matching; temperature rescales sensitivity. These correspondences provide a compact language for expressing how model behavior changes across training and decoding settings.

**Relation to “stochastic parrots.”** Bender et al. (2021) introduced the metaphor of “stochastic parrots” to highlight concerns about large-scale next-token prediction, including risks of bias amplification, ecological costs, and questions about meaning. We agree that these are serious considerations for responsible deployment. At the same time, our results clarify that modern LLMs differ formally from literal shufflers of surface statistics. Whereas finite-order  $n$ -gram generators produce outputs by local concatenation with no global allocation principle, contemporary LLMs are trained by objectives that yield allocation dynamics consistent with the matching law. From a behavior-analytic perspective, this constitutes a kind of grounding: models adapt allocations in proportion to contingencies, in the same sense that organisms adapt behavior to experiential reinforcement. Thus, while LLMs do not possess symbolic reference in the classical semantic sense, they are not ungrounded; their “semantics” are embodied in learned allocations. This distinction

| Row                                                  | Research Questions                                      | Minimal Tests / Stretch Goals                                                                                                                                                                         |
|------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operant $\leftrightarrow$ MLE (strict)               | Context granularity where strict matching holds?        | <i>Minimal:</i> Log-odds regressions on held-out corpora; slope $\approx 1$ , intercept $\approx 0$ .<br><i>Stretch:</i> Multi-token alternatives; compositional continuations; multilingual corpora. |
| Generalized matching $\leftrightarrow$ KL-alignment  | Map $\beta$ to sensitivity across datasets/models?      | <i>Minimal:</i> DPO with $\beta$ sweep; fit slope/bias from log-odds shifts.<br><i>Stretch:</i> PPO-style RLHF; compare to animal sensitivity estimates.                                              |
| Inference temperature                                | How stable is $1/T$ scaling across prompts/models?      | <i>Minimal:</i> Temperature sweep; recover slopes $1/T$ .<br><i>Stretch:</i> Adaptive temperature; task-conditioned sensitivity.                                                                      |
| Classical $\leftrightarrow$ supervised               | Do cueoutcome learning curves match associative models? | <i>Minimal:</i> Train small nets on cueoutcome pairs; fit RescorlaWagner-like curves.<br><i>Stretch:</i> Blocking/overshadowing analogs in pretraining slices.                                        |
| Hebbian $\leftrightarrow$ self-supervised reps       | Are updates consistent with correlation-driven rules?   | <i>Minimal:</i> Track embedding changes under masked-LM objectives.<br><i>Stretch:</i> Compare with neuro data (STDP-like signatures).                                                                |
| Observational $\leftrightarrow$ in-context           | How do demonstrations shape one-shot generalization?    | <i>Minimal:</i> Prompt with examples; measure performance vs. number of demos.<br><i>Stretch:</i> Sample-efficiency laws; compare to animal imitation.                                                |
| Memory constraints $\leftrightarrow$ context windows | How does allocation fidelity scale with window length?  | <i>Minimal:</i> Strict matching vs. window size.<br><i>Stretch:</i> External retrieval as extended working memory; quantify gains.                                                                    |

Table 1: Expanded research program connecting behavior-analytic constructs to LLM mechanisms. Report sensitivity (slope), bias (intercept),  $R^2$ , and confidence intervals; include recency and capacity bounds where relevant.

separates them sharply from genuine parroting baselines (see Table 4 and Appendix A for code and examples).

**On stochastic parrots and plagiarism.** Saying that a model repeats its training data redescribes strict matching: allocations follow contingencies. The scientifically interesting question is how and when models *deviate* bias and undermatching quantify the impact of regularization, data

mixture, and architectural limits. These deviations are diagnostic of learning structure rather than evidence of mere parroting.

## What Kind of Grounding Do LLMs Provide?

A central critique associated with the “stochastic parrot” metaphor is that large-scale next-token prediction lacks semantic grounding. This assumes that meaning requires symbolic reference to an external world model. From a behavior-analytic perspective, however, grounding is not defined by internal representational structures but by lawful adaptation of responses to contingencies. In Skinners formulation, behavior is “selected by consequences”; allocation is grounded insofar as it tracks reinforcement in the environment.

Our formal results show that LLMs instantiate precisely this kind of allocation-grounding. During pretraining, maximum likelihood estimation drives probabilities to match the distribution of linguistic contingencies in the corpus. During alignment, KL-regularized objectives tilt allocations in interpretable ways, analogous to sensitivity and bias in generalized matching. At inference, decoding parameters such as temperature rescale these allocations systematically. In each case, allocation is not arbitrary imitation but an adaptive response to experienced contingencies.

Thus, while LLMs do not possess semantic grounding in the classical symbolic sense, they are not ungrounded. Their semantics are *operant semantics*: meanings emerge as stable allocation patterns shaped by exposure and feedback. This reframing clarifies why the “stochastic parrot” label, though apt for  $n$ -gram generators, obscures what modern models actually achieve. LLMs are grounded in their experience, much as organisms are grounded in reinforcement histories, and this grounding can be formalized and tested using the matching law correspondences. The broader empirical program we outline (Table 1) specifies concrete experiments that make these allocation-based grounding claims testable across scales, contexts, and training regimes.

Importantly, these correspondences are not tied to obsolete constructs. Matching law analysis is still a cornerstone of contemporary behavioral research and application (Kronfli et al., 2021; Rajagopalan et al., 2023; Anderson, 2024; Mastermind Behavior, 2025), demonstrating that the laws we connect to LLM objectives remain both empirically and clinically robust.

**On generative linguistics.** Our claims do not reduce syntax or linguistic competence to reinforcement. Following Marrs levels of analysis (Marr, 1982), matchinglaw correspondences operate at the level of allocation dynamics and complement structural accounts. They explain *how* responses are allocated under contingencies, not *which* structures are licensed by a grammar.

**Ethics and normativity.** The framework is descriptive, not prescriptive. It does not evaluate whether the contingencies used to train LLMs are desirable; rather, it clarifies what those contingencies are and how they shape behavior. Ethical analyses remain essential, but they benefit from models that make testable predictions instead of relying on metaphors.

As a roadmap for future work, Table 3 maps other psychological learning paradigms to machinelearning counterparts (e.g., classical conditioning to supervised learning; Hebbian learning to selfsupervised representation learning; observational learning to imitation and incontext learning). This suggests a broader comparative science spanning organisms and machines.

Our analysis has emphasized the correspondence between the matching law and allocation in large language models. Yet Skinner himself, while committed to a black-box analytic stance, did not deny that internal mechanisms constrain what an organism can learn. An organism cannot allocate behavior according to reinforcement contingencies that exceed its perceptual or memory capacity. For this reason, even within operant psychology, working memory limitations and reinforcement delays were recognized as important moderators of matching performance (Baum, 1974; Shimp, 1976).

An analogous constraint appears in large language models. The context window defines the effective working memory of the model: the span of prior tokens on which allocation decisions can be conditioned. Just as pigeons undermatch when reinforcement histories extend beyond their retention interval, LLMs fail to reproduce long-range statistical dependencies that exceed their context window. Both systems therefore approximate the matching law only within the boundaries of their capacity to track contingencies.

This analogy can be taken further. Recency effects in human memory parallel the recency bias in transformer attention mechanisms. Similarly, expansion of context windows in newer architectures (e.g., from 2k to 128k tokens) yields closer adherence to corpus statistics across longer spans, just as enhanced working memory capacity in humans supports more complex contingency tracking. Where the correspondence breaks down is also informative: biological systems integrate multiple memory stores (working, episodic, semantic), whereas LLMs rely on a single bounded attention buffer.

The implication is that memory capacity acts as a structural moderator of matching-law correspondence in both animals and machines. Allocation follows reinforcement contingencies to the extent that those contingencies can be represented within the available memory system. This extension highlights that behaviorist principles are not invalidated by internal constraints, but rather shaped by them.

Just as an organism’s learning is inseparable from the totality of its perceptual gestalt, an LLM’s learning is inseparable from the gestalt of its training corpus. This corpus constitutes a composite, a pastiche gestalt, drawn from the described and imagined experiences of countless humans.

**Allocation- vs. Gestalt-Grounding.** We distinguish between two complementary senses of grounding. *Allocation-grounding* refers to the generic, substrate-independent form of grounding: behavior (or predictions) is grounded insofar as its allocations track contingencies, a relation captured formally by the matching law and its extensions. In contrast, *gestalt-grounding* refers to the particularized substrate or experiential channel through which contingencies are encountered. For organisms, the gestalt is perceptual and sensory; for LLMs, it is textual and corpus-based, a “pastiche gestalt” assembled from fragments of human experience. Allocation-grounding provides the general law, while gestalt-grounding specifies the contextual medium in which such allocations unfold. For LLMs, *gestalt-grounding* would refer to the models’ allocations being grounded in a distributed human gestalt, even if it lacks direct perception.

## 8 Conclusions

This work establishes formal correspondences between principles of operant behavior analysis and the optimization dynamics of large language models. We showed that (i) maximum likelihood estimation instantiates *strict matching* between model and corpus distributions, (ii) KL-regularized preference optimization instantiates *generalized matching* with interpretable sensitivity and bias parameters, and (iii) temperature scaling at inference respecifies sensitivity by a factor of  $1/T$ . These results provide a substrate-independent account of allocation that unites behavioral laws with modern machine learning objectives.

In doing so, we clarify why the “stochastic parrot” metaphor, while literally apt for finite-order  $n$ -gram shufflers, obscures the allocation dynamics of contemporary LLMs. From a behavior-analytic perspective, LLMs are not ungrounded mimics but adaptive systems whose responses are *allocation-grounded* in their experiential contingencies. In parallel, their corpus functions as a *gestalt-like substrate*, a pastiche of human experience that provides the conditions under which

| Property           | Finite-order $n$ -gram generator (e.g., dissociated-press, <i>Travesty</i> )                            | Large Language Model (Transformer-based)                                                                                              |
|--------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| State dependence   | Fixed-order Markov chain: next token depends only on $n - 1$ previous tokens.                           | Self-attention over extended context window (thousands of tokens), enabling long-range dependencies.                                  |
| Training objective | None beyond frequency counts of substrings.                                                             | Maximum likelihood estimation via cross-entropy minimization (equivalently, KL minimization to corpus distribution).                  |
| Allocation law     | None: transitions are raw empirical frequencies; no principled allocation law applies.                  | Formal correspondences to strict and generalized matching; allocation of probability mass follows reinforcement-like contingencies.   |
| Generalization     | None: fails catastrophically on out-of-vocabulary sequences; cannot interpolate unseen patterns.        | Learns distributed representations; generalizes to unseen contexts via embeddings and attention.                                      |
| Control parameters | $n$ (order of Markov chain).                                                                            | Temperature, top- $k$ , nucleus sampling, etc., which directly rescale sensitivity and variability.                                   |
| Critique fit       | “Stochastic parrot” metaphor is literally accurate: purely parrots substrings without deeper structure. | Better described as <i>allocation matchers</i> obeying cross-domain laws of choice; metaphor of “parrot” obscures these regularities. |

Table 2: Contrasting classical  $n$ -gram text generators with modern large language models. The “stochastic parrot” metaphor applies literally to the former, while LLMs instantiate allocation principles that extend beyond surface shuffling.

adaptation occurs. This form of *gestalt-grounding* complements allocation-grounding: together they frame LLM behavior as grounded in both the dynamics of reinforcement analogs and the structure of the corpus gestalt.

Beyond theory, the correspondences yield testable predictions. The expanded research program (Table 1) specifies minimal and stretch experiments that operationalize allocation-based and gestalt-based grounding across scales, alignment regimes, and inference settings. These include systematic tests of matching-law adherence under different context windows, preference weights, and decoding parameters. Such experiments move the conversation from metaphor to measurement, allowing deviations to be interpreted diagnostically as bias or undermatching rather than as evidence of vacuity.

More broadly, our results contribute to a common framework for analyzing learned systems. Just as behavior analysis abstracts from neural substrates, our correspondence abstracts from model architecture. Allocation and gestalt laws appear to span organisms, artificial agents, and text models alike. Recognizing this continuity not only answers some critiques but also opens

avenues for comparative science: grounding is not binary but graded, observable in the degree to which responses lawfully adapt to contingencies within a given gestalt.

**Summary.** LLMs do not merely parrot surface forms. They instantiate allocation and gestalt principles that connect them to one of psychology’s most robust empirical laws. This reframing offers both a descriptive account of present models and a foundation for empirical research programs that probe their capabilities and limits.

The same allocation laws that structured mid-20th century analyses of pigeon behavior (Herrnstein, 1961; Baum, 1974) are still evident in contemporary applications and neuroscience (Kronfli et al., 2021; Rajagopalan et al., 2023; Anderson, 2024; Mastermind Behavior, 2025). By extending these laws to LLMs, through both allocation-grounding and gestalt-grounding, we situate our results in a lineage that spans from classic operant studies to today’s most advanced models of learning.

## Limitations and Future Work

Our analysis is deliberately narrow: it formalizes correspondences between LLM objectives and the matching law. This scope has several limitations. First, we do not claim to address all aspects of human language use, such as intentional semantics, pragmatic inference, or embodied interaction. From a behavior-analytic perspective, our claims are descriptive of allocation, not explanatory of higher-order meaning.

Second, the framework abstracts away from architectural details. While this substrate-independence is a strength, it also means we do not model how attention, embeddings, or scaling laws contribute to allocation fidelity. Future work should integrate mechanistic analyses of representation with the allocation perspective.

Third, our proofs assume realizable conditions and infinite data. In practice, finite corpora, limited capacity, and regularization induce deviations (undermatching, bias) that must be measured empirically. The research program outlined in Table 1 specifies minimal experiments, but scaling to frontier models will require substantial computational resources.

Finally, our treatment of grounding is behavior-analytic: allocations are grounded insofar as they adapt to contingencies. Critics may prefer symbolic or referential notions of semantics. Bridging these accounts remains an open challenge. Future work should explore how allocation-grounding interacts with other forms of learning, such as classical conditioning, Hebbian plasticity, or in-context generalization, and whether hybrid frameworks can reconcile behavioral and symbolic perspectives.

In sum, the present work reframes LLM behavior through the matching law lens, but it is only a first step. Extending this framework to broader settings, testing it across models and scales, and connecting it to semantic theories are important tasks for future research.

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## 10 Paradigm Mapping

| Learning Paradigm                   | Biological Example                         | Machine-Learning Counterpart                             |
|-------------------------------------|--------------------------------------------|----------------------------------------------------------|
| Operant conditioning                | Matching law in pigeons (Herrnstein, 1961) | Maximum likelihood training (strict matching)            |
| Reinforcement with bias/sensitivity | Generalized matching (Baum, 1974)          | KL-regularized RLHF/DPO (sensitivity parameter $\beta$ ) |
| Inference temperature               | Response variability scaling               | Logit temperature scaling in sampling                    |
| Classical conditioning              | Pavlovs dogs (stimulusresponse)            | Supervised learning (inputoutput mapping)                |
| Hebbian learning                    | Synaptic plasticity                        | Self-supervised representation learning                  |
| Observational learning              | Imitation in primates                      | In-context learning / few-shot prompting                 |
| Memory constraints                  | Working memory span, recency effects       | Context window size, recency bias in attention           |

Table 3: Mapping of psychological learning paradigms to machine-learning counterparts, extended to include memory constraints.

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## A Additional Comparisons

### A.1 N-gram Generators vs. Large Language Models

The phrase “stochastic parrot” (Bender et al., 2021) is in some sense most literally applicable to finite-order  $n$ -gram text generators such as Emacs `dissociated-press` or the *Travesty* program published in *Byte Magazine* (1984). These programs produce text by shuffling local substrings according to observed  $n$ -gram frequencies, with no global objective or allocation principle. By contrast, modern large language models are trained by maximum likelihood to minimize cross-entropy, yielding probability allocations consistent with the matching law and its generalizations.

Table 4 contrasts the two approaches. The comparison clarifies that while the “parrot” metaphor fits early  $n$ -gram systems literally, LLMs instantiate optimization principles that link them to quantitative laws of behavior, even though questions of semantic grounding and bias remain important open issues.

| Property           | Finite-order $n$ -gram generator (e.g., <code>dissociated-press</code> , <i>Travesty</i> )              | Large Language Model (Transformer-based)                                                                                              |
|--------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| State dependence   | Fixed-order Markov chain: next token depends only on $n - 1$ previous tokens.                           | Self-attention over extended context window (thousands of tokens), enabling long-range dependencies.                                  |
| Training objective | None beyond frequency counts of substrings.                                                             | Maximum likelihood estimation via cross-entropy minimization (equivalently, KL minimization to corpus distribution).                  |
| Allocation law     | None: transitions are raw empirical frequencies; no principled allocation law applies.                  | Formal correspondences to strict and generalized matching; allocation of probability mass follows reinforcement-like contingencies.   |
| Generalization     | None: fails catastrophically on out-of-vocabulary sequences; cannot interpolate unseen patterns.        | Learns distributed representations; generalizes to unseen contexts via embeddings and attention.                                      |
| Control parameters | $n$ (order of Markov chain).                                                                            | Temperature, top- $k$ , nucleus sampling, etc., which directly rescale sensitivity and variability.                                   |
| Critique fit       | “Stochastic parrot” metaphor is literally accurate: purely parrots substrings without deeper structure. | Better described as <i>allocation matchers</i> obeying cross-domain laws of choice; metaphor of “parrot” obscures these regularities. |

Table 4: Contrasting classical  $n$ -gram text generators with modern large language models. The “stochastic parrot” metaphor applies literally to the former, while LLMs instantiate allocation principles that extend beyond surface shuffling.

## Illustrative Code: $n$ -gram Text Generator

For completeness, we include a simple Python implementation of an  $n$ -gram generator (“travesty” program). It builds a frequency table from a corpus and then produces text by sampling next tokens solely from the observed local  $n$ -gram statistics. No global optimization or allocation principle is involved.

```
import random
from collections import defaultdict, Counter

def train_ngram(corpus, n=3):
    """Train an n-gram frequency table from a list of tokens."""
    model = defaultdict(Counter)
    for i in range(len(corpus)-n):
        context = tuple(corpus[i:i+n-1])
        next_tok = corpus[i+n-1]
        model[context][next_tok] += 1
    return model

def generate_text(model, length=50, seed=None):
    """Generate text by stochastic concatenation of n-grams."""
    context = random.choice(list(model.keys()))
    output = list(context)
    for _ in range(length-len(context)):
        next_tok = random.choices(
            list(model[context].keys()),
            weights=model[context].values()
        )[0]
        output.append(next_tok)
        context = tuple(output[-(len(context)):])
        if context not in model:
            context = random.choice(list(model.keys()))
    return " ".join(output)

# Example usage
tokens = "the cat sat on the mat and the cat lay on the rug".split()
model = train_ngram(tokens, n=3)
print(generate_text(model, length=20))
```

This generator can reproduce local phrases but fails rapidly at longer spans: it lacks representations, optimization principles, or allocation laws. In contrast, large language models trained by maximum likelihood estimation implement allocation dynamics consistent with the matching law.

## Methodological Note on Output Generation

**N-gram generator.** We used the simple Python implementation shown above with order  $n=3$ . The training corpus was a short English sentence repeated with minor variations: “The cat sat on the mat and the cat lay on the rug.” The generator was run for 50 tokens with a randomly chosen seed context. As expected, the output consists of repetitive concatenations of local substrings, with no global coherence.

**LLM output.** For comparison, we used a transformer-based large language model (e.g., GPT-3.5 or GPT-4) with the same seed prompt: “The cat sat on the mat”. Inference was performed with temperature  $T=1.0$ , top- $k=50$ , and nucleus sampling  $p=0.95$ . The model generated approximately 50 tokens of continuation. The sample shown in Figure ?? is representative of coherent continuation under these settings.

**Interpretation.** This side-by-side comparison illustrates the rhetorical point: the metaphor of “stochastic parrots” applies literally to finite-order  $n$ -gram shufflers, whereas LLMs instantiate allocation laws consistent with maximum-likelihood and matching-law principles. This does not by itself resolve questions of semantic grounding or bias, but it demonstrates that the mechanisms underlying LLM outputs are formally distinct from mere substring shuffling.